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BADJI MOKHTAR UNIVERSITY -ANNABA-
FACULTY OF MEDICINE
DEPARTMENT OF MEDICINE
LABORATORY OF MEDICAL AND SURGICAL ANATOMY

Introduction to human anatomy

Course notes intended for first-year medical students.

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Course outline

Introduction

1- Definitions
2- History of anatomy
3- Branches of anatomy
4- Study methods

Bibliography

Introduction

Anatomy is an ancient science dating back to antiquity, since the word anatomy comes from the Greek ανατομία and the Latin anatomia, both meaning dissection. It is thesciencewhich aims to study andawarenessof the parts which make up living organisms, their form, their dimensions, their relationships, their structure, etc..,either by means of dissection, or by any other method of investigation and research .

1- Definitions

- From the Greek: anatomia or anatome, meaning "to cut through" or "to dissect".
- Paturet (1951): it is the "science whose object is the study of the constitution of organized beings".
- Rouvière: it's the "science of body structures".
- From Ribet (1961): it is the "science which has as its object the study of the form, the reciprocal relationships and the final structure of the organs of organized beings, among them: man".
- **Importance of anatomy :**
  - Ibn Rushd: "Whoever studies anatomy increases his faith in the omnipotence and unity of Almighty God."
  - Vesalius (1543): anatomy "should absolutely be considered as the only solid basis of the whole art of medicine."
  - Paturet (1951): "The study of human anatomy is of considerable practical interest, because it is of interest not only to the physician and the surgeon, but also to the artist, the biologist, the physiologist, the anthropologist; but it is above all the basic science, the fundamental science of medical studies, the one on which the study of physiology, clinical practice and surgical technique is based."

2- History of anatomy

2.1. The beginnings of anatomy

- 3000 BC: first anatomical descriptions on papyrus (heart, blood, liver...), moreover the mummification of bodies testifies to their knowledge of anatomy.
- Hippocrates (460–377 BC): taught human anatomy in Greece.
- Aristotle (384–322 BC): He is the founder of comparative anatomy; he used the term "anatomy" for the first time.
- Claudius Galen (131–201 AD), “prince of physicians”: taught anatomy and wrote on the subject. He conducted his research on monkeys.

2.2. Anatomy among Muslims (from the 10th to the 13th century)

2.2.1.1. Ibn Al-Haitham (965-1040): He was the first to describe the anatomy of the eye; he is known for his famous research on lenses.

2.2.1.2. Ibn Sina (Avicenna) (980-1037): used corpses from the battlefields to dissect, and wrote "Canon of Medicine".

2.2.1.3. Ibn Rushd (Averroes) (1126-1198): wrote "Anatomy of the Organs"

2.2.1.4. Ibn an-nafis (1213-1288): reveals the first description of pulmonary circulation, and writes "comments on the anatomy of the Canon of Ibn Sina".

2.3. Renaissance Anatomy (15th to 16th century)

2.3.1. Jacques Dubois (also known as Sylvius) (1478-1555): described the middle cerebral artery.

2.3.2. Gabriel of Fallopia: describes the uterine tube and the facial nerve.

2.3.3. André Vésale (1514-1564) :

- Professor of anatomy.
- Reference work of « Humani Corporis Fabrica » published in 1543.
- First public dissection.

2.4. 17th Century Anatomy

2.4.1. William Harvey (1578-1657): rediscovered and demonstrated pulmonary circulation.
2.4.2. Malpighi: discovers capillary vessels.
2.4.3. Steno: demonstrates cardiac muscle contraction.
2.4.4. Aselli, Pecquet: discover and demonstrate lymphatic circulation.

2.5. 18th Century Anatomy

2.5.1. This period saw the birth of comparative anatomy, anthropology, and biology.
2.5.2. The foundation of pathological anatomy by Morgagni (1682-1771).
2.5.3. The appearance of dissection amphitheaters and natural history galleries throughout Europe.

2.6. Anatomy in the 19th century

2.6.1. The period in histology when the cell theory was demonstrated.
2.6.2. The development of topographic and surgical anatomy for the safety of amputations.
2.6.3. Corti, Pacini, Golgi: focus their work on neuroanatomy.
2.6.4. 1819-1899: appearance of various tools allowing the study of anatomy on living subjects (stethoscope, otoscope, ophthalmoscope, laryngoscope, gastroscope, cystoscope, bronchoscope).
2.6.5. Around 1890: use of formaldehyde as a fixative.
2.6.6. Conrad Roëtgen (1845-1923): discovered X-rays in 1895.

2.7. Current anatomy

2.7.1. Anatomy is becoming primarily applied, for medicine, surgery, imaging and teaching.
2.7.2. 1993 : Visible Man Project, a 3D virtual man that allows us to see real human anatomy, after the dissection of a condemned man into 1800 thin slices.

3- Branches of anatomy

3.1. Descriptive or explanatory anatomy

Descriptive anatomy is the analytical study of the morphology of separate organs (shape, dimensions, weight, color, consistency, constitution, internal structure…).
It represents the basic study in anatomy.

3.2. Topographic or regional anatomy

Topographical anatomy is the study of the location and relationships of organs within the same anatomical region.
This is the basic study of surgery.

3.3. Functional Anatomy

Functional anatomy is the study of the function of organs and their relationship to morphology.

3.4. Systemic Anatomy

Systemic anatomy is the study of the body's systems and organs.

3.5. Radiological Anatomy

Radiological anatomy is the study of morphology using imaging techniques (X-ray, CT scan, MRI, scintigraphy, etc.).

3.6. Medical-surgical, clinical, or applied anatomy

It is anatomy applied to the clinic.

3.7. Comparative Anatomy

Comparative anatomy is the study of the relationships existing between homologous structures of all animals including humans, either between individuals (ontogeny) or between species (phylogeny).

3.8. Anthropological Anatomy

Anthropological anatomy is the study of morphological variations in human races.

3.9. Anatomy of Development

The study of the morphological transformations of the organism from fertilization to adulthood. It encompasses prenatal development (embryology, fetology), postnatal development (growth) and teratology (congenital malformations).

3.10. Pathological Anatomy

The study of alterations in anatomical structures caused by diseases.

3.11. Microscopic or structural anatomy

The microscopic study of the structure of cells (cytology) and tissues (histology).

3.12. Artistic or plastic anatomy or forms

The study of the external forms of the human body. It is intended for sculptors, engravers and painters.

4. Study Methods

4.1. Dissection
4.2. Plastination
4.3. Imaging (radiology, ultrasound, CT scan, MRI, arteriography, IVU, scintigraphy…)
4.4. Endoscopy (bronchoscopy, rectoscopy)
4.5. Surgery,
4.6. Microscopy.

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